

Bharatiya Antariksh Hackathon 2026



Problem Statement: *Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning*

Can you discover hidden lunar ice deposits and design India's future robotic exploration strategy around them?

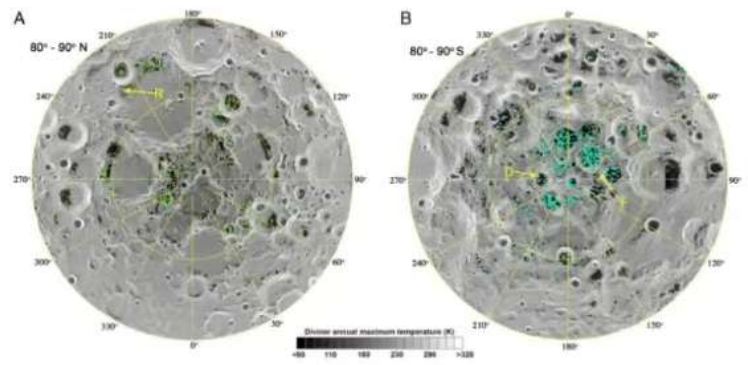
I have Chandrayaan-2 radar and imagery data. I need to find subsurface ice, select a landing site, plan a rover route, and estimate ice volume.

DFSAR + OHRC Data
↓
Detect Subsurface Ice
↓
Select Landing Site
↓
Design Rover Traverse
↓
Estimate Ice Volume (0–5 m)

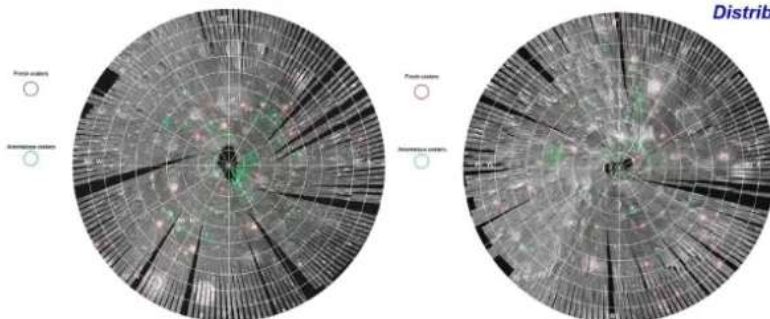
What students should feel proud of:

- *Where exactly is subsurface ice located?*
- *How can ice be distinguished from rocky terrain?*
- *Which sites are safe for landing?*
- *How can a rover reach the target efficiently?*
- *How much ice may actually exist?*

Evidence for Lunar Polar Ice — But Are We Done Yet?

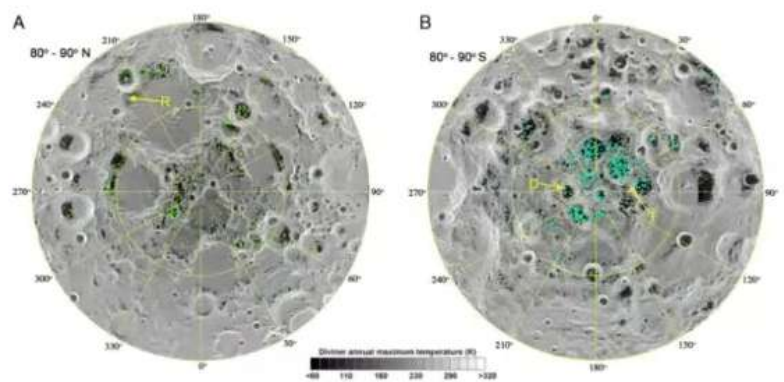


Distribution of ice-bearing pixels constrained by M3, LOLA and LAMP (Li et al., 2018)

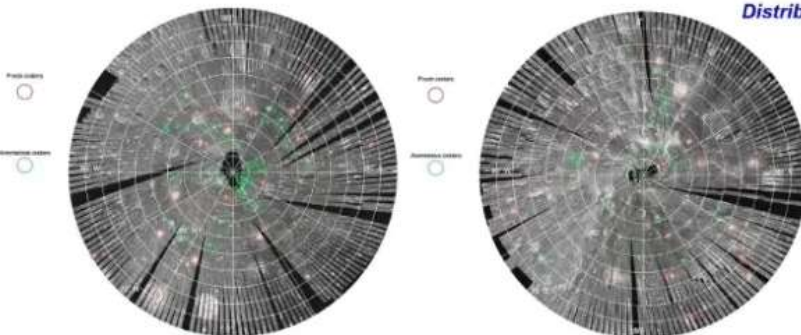


Location maps for the anomalous craters in north and south polar regions that have higher than average CPRs (modified from Spudis et al., 2013)

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Key gap areas in polar ice detection:

- Ambiguity in radar-based detection
- Poor constraints on ice depth and volume
- Incomplete understanding of micro environments
- Ice formation and delivery mechanisms
- Surface vs. subsurface discrimination
- Spatial heterogeneity

npj Space Exploration has APC waivers available that can be allocated upon acceptance on an *ad-hoc* basis. For additional information, contact the Journal Publisher, [Jialiang Cai](#).

Article Open access Published: 06 May 2026

Subsurface ice in doubly shadowed craters as revealed by Chandrayaan-2 dual frequency synthetic aperture radar

Rishitosh K. Sinha, Rajiv R. Bharti, Kinsuk Acharyya, Sanjay K. Mishra, Neeraj Srivastava & Anil Bhardwaj

npj Space Exploration 2, Article number: 22 (2026) | Cite this article

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Abstract

Doubly shadowed craters at the lunar poles provide ideal environments for preserving subsurface ice. Located within permanently shadowed regions, their raised rims additionally block scattered light and thermal emission from nearby illuminated surfaces, allowing crater interior temperatures to reach around 25 K. Using full-polarimetric L- and S-band observations from Chandrayaan-2's Dual Frequency Synthetic Aperture Radar, we investigate the presence of subsurface ice in nine doubly shadowed craters located within Faustini, Haworth, and Shoemaker craters. Four craters exhibit circular polarization ratio values greater than 1

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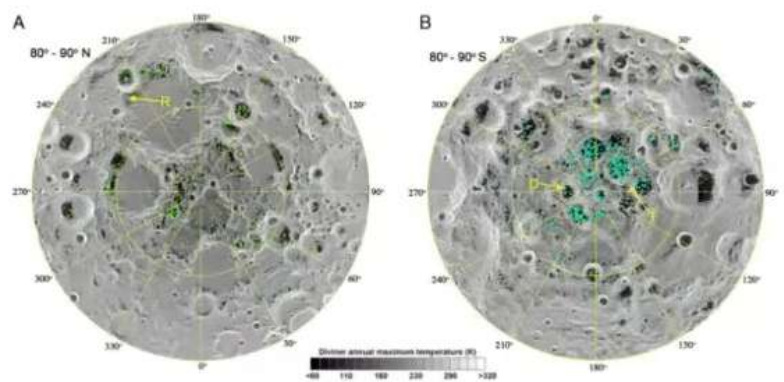
Associated content

Collection

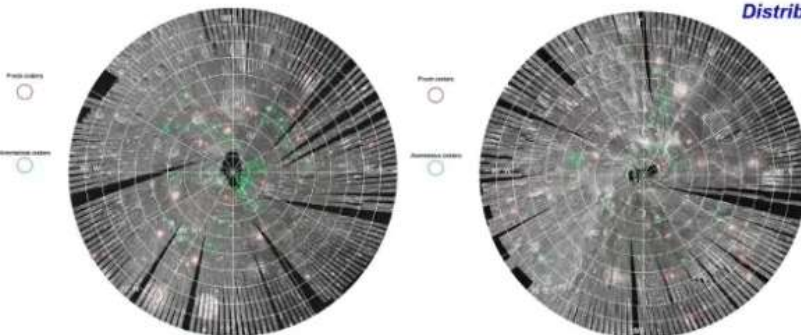
Lunar Surface Landscape and Shallow Structure: Craters and Lava Tubes

Sections	Figures	References
Abstract		
Introduction		
Results		
Discussions		
Methods		
Data availability		
References		

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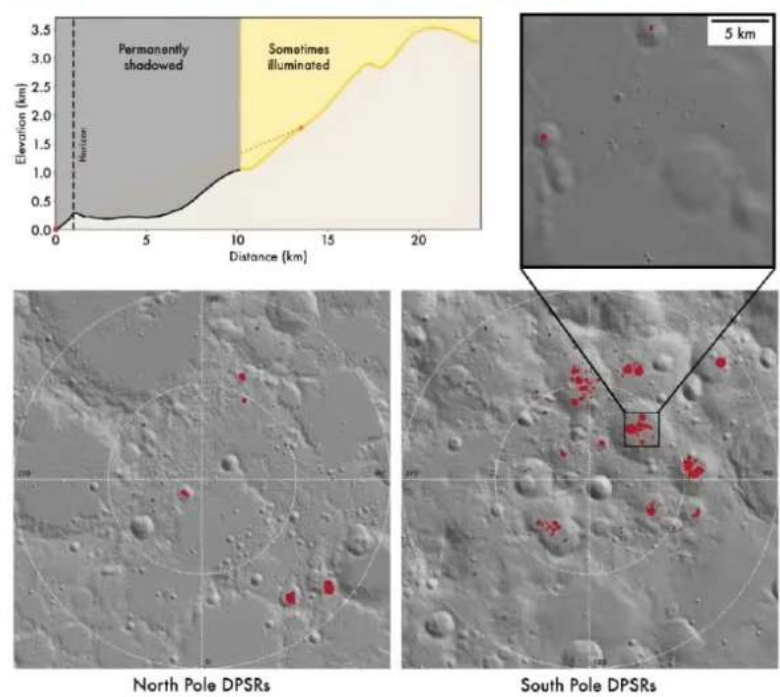


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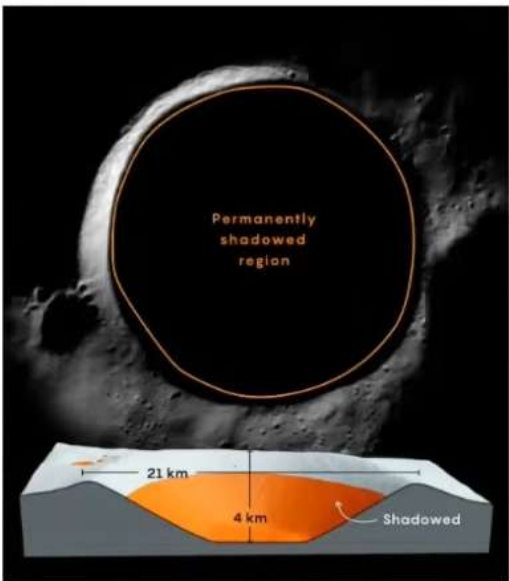
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What are 'Doubly Shadowed' craters?



A point is doubly shadowed if it is permanently shadowed and has no direct line of sight to any non-permanently shadowed surfaces.



The north-south axis of the Moon is nearly perpendicular to the plane of the solar system. The rims of craters block sunlight from reaching their depths.

Merrill Sherman/Quanta Magazine; Source: NASA

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Associated content

Collection

Lunar Surface Landscape and Shallow Structure: Craters and Lava Tubes

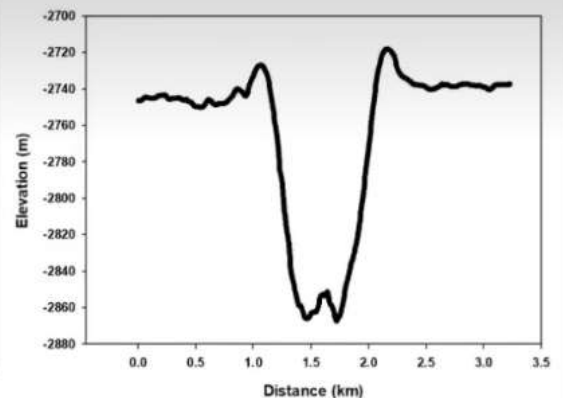
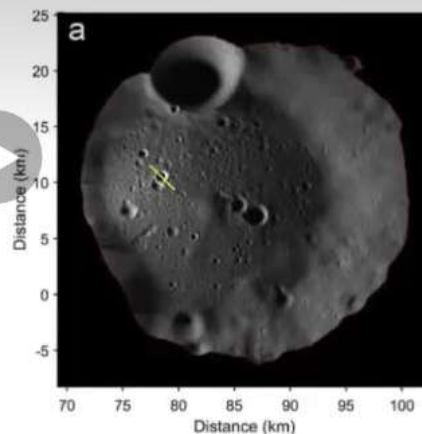
Sections Figures References

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The rim of the doubly shadowed crater shields its interior from scattered light and thermal emission from the periodically illuminated rim of the larger crater. These double permanently shadowed regions (DPSRs) are among the coldest places in the inner solar system, with theoretical temperatures approaching 25 K—the minimum possible surface temperature on the Moon.

Table 1
Size and Location of the Largest Doubly Shadowed Regions on the Moon

Area (km ²)	D _{eq} (m)	Latitude (°)	Longitude (°)	Lunar Features
0.274	590.219	85.552	49.426	Fibiger, Byrd D, Nansen F Crater
0.262	577.461	-88.275	38.992	Shoemaker Crater
0.242	555.204	-85.362	48.430	Nobile Crater
0.220	529.858	85.636	50.220	Fibiger, Byrd D, Nansen F Crater
0.213	521.135	-86.870	24.335	Haworth, Shoemaker Crater
0.201	505.509	-87.390	82.312	Faustini Crater
0.187	488.211	86.035	36.598	Fibiger Crater
0.182	481.118	-86.940	19.223	Haworth, Shoemaker Crater
0.176	473.919	-87.114	85.381	Faustini Crater
0.167	461.671	86.113	36.769	Fibiger Crater
0.152	440.068	-88.171	117.626	Slater Crater
0.126	400.535	-87.915	-10.942	Haworth Crater
0.121	391.858	-87.381	-6.105	Haworth Crater
0.116	384.478	87.086	158.568	Bosch Crater
0.114	381.486	-87.841	40.078	Shoemaker Crater
0.111	375.430	-87.436	80.655	Faustini Crater
0.076	312.094	-87.310	86.333	Faustini Crater
0.071	300.878	-88.531	-141.567	Sverdrup Crater
0.070	298.967	-88.049	41.710	Shoemaker Crater
0.068	295.109	89.398	-54.699	Hinschwood Crater
0.063	283.221	-89.053	-45.939	de Gerlache, Haworth Crater
0.061	279.146	-87.138	-8.665	Haworth Crater
0.058	270.811	87.621	151.865	Bosch Crater



An example of a doubly shadowed crater within Faustini PSR

ISRO Website Coverage

भारतीय अंतरिक्ष अनुसंधान संगठन, अंतरिक्ष विभाग
Indian Space Research Organisation, Department of Space
भारत सरकार / Government of India

Home About Activities Services Programmes Resources Engagements

Chandrayaan-2 Dual Frequency Synthetic Aperture Radar (DFSAR) Observations Reveal Subsurface Ice in Lunar South Polar Regions

Home / Chandrayaan-2 Dual Frequency Synthetic Aperture Radar (DFSAR) Observations Reveal Subsurface Ice in Lunar South Polar Regions

May 27, 2026

A detailed investigation of sub-surface ice in the lunar South Polar Region using observations from the Chandrayaan-2 Dual Frequency Synthetic Aperture Radar (DFSAR) have been carried out by the scientists from Physical Research Laboratory (PRL), Ahmedabad.

The Dual Frequency Synthetic Aperture Radar (DFSAR) onboard Chandrayaan-2 orbiter is a microwave imaging instrument in L- and S-band frequencies and the first fully polarimetric SAR to study the Moon. The study focuses on 'doubly shadowed craters,' which are special craters located inside permanently shadowed regions (PSRs) of the Moon. Due to continuous shielding from sunlight and thermal radiation, these regions remain extremely cold (temperatures ~29K) and are considered favorable locations for preserving water/ice over long geological timescales.

Using advanced radar polarimetric analysis, scientists identified radar signatures consistent with the possible presence of subsurface ice beneath the floors of four doubly shadowed craters in the lunar South Polar Region. The study proposes a refined radar-based approach for identifying subsurface ice, where Circular Polarization Ratio (CPR) values greater than 1 together with Degree of Polarization (DOP) values less than 0.13 indicate volumetric scattering potentially associated with subsurface ice. DOP is a radar polarimetric parameter that measures how much of the scattered radar signal retains its original polarization state after interacting with the surface or subsurface material. This approach helps distinguish genuine ice signatures from radar signals produced by rough rocky terrain.

Among the investigated craters, one crater of 1.1 km diameter within Faustini crater shows particularly strong evidence of subsurface ice (Figure 1), supported by both radar observations (Figure 2) and distinctive lobate-rim morphological characteristics. A lobate-rim morphology refers to flow-like or lobed appearance, suggesting the impact may have penetrated subsurface ice, producing the observed lobate-rim crater.



Doubly shadowed crater with lobate-rim

Chandrayaan-2 Dual Frequency Synthetic Aperture Radar (DFSAR) Observations Reveal Subsurface Ice in Lunar South Polar Regions (Date: May 27, 2026)

THE TIMES OF INDIA

CHANDRAYAAN-2 FINDS ICE BURIED ON LUNAR SOUTH POLE

PRL study sheds new light on Moon's icy secrets

Parth Bharti
@timesofindia.com

Ahmedabad: There is water in the form of ice crystals beneath the surface of the Moon in areas where direct or indirect sunlight or heat does not reach, indicates research by a team from the Physical Research Laboratory (PRL) in Ahmedabad. The find is based on strong radar evidence from data of Chandrayaan-2 mission of ISRO.

The radar signatures from 'doubly shadowed' craters in the South Polar region provided new evidence on how water — the lifeline for future human plans to use the Earth's natural satellite for space exploration — is retained for millions of years in absence of atmosphere and extreme temperature fluctuations.

The study 'Subsurface ice in doubly shadowed craters as revealed by Chandrayaan-2 dual frequency synthetic aperture radar' by Rohitash K. Raut, Rajiv R. Bharti, Kinshuk Acharyya, Sanjay K. Mishra, Naveen Srivastava, and Anil Bhattacharya, was published recently in the Springer Nature NPJ journal 'Space Exploration'.

Sucha, lead author, said deep impact craters are located in the permanently shadowed region (PSR) that does not receive direct sunlight. "Moreover, its raised rim additionally blocks scattered light and thermal emission from nearby illuminated surfaces. Thermal models suggest interior temperature reaches around 25 kelvin (approx. -248°C) which is lower than even some outer planets," he said.



HOW DISCOVERY WAS MADE
PRL team analysed the data of double frequency synthetic aperture radar (DFSAR) of the Chandrayaan-2 orbiter and found ice beneath the lunar surface.

The team used full-polarimetric L- and S-band observations from Chandrayaan-2's Dual Frequency Synthetic Aperture Radar (DFSAR) to investigate nine doubly shadowed craters within three craters: Faustini, Bhattacharya, and Shumaker.

The radar scan showed potential signatures for ice beneath the surface of four of these nine craters. Team members said they employed circular polarisation ratio (CPR), noting that radar waves undergoing volumetric scattering within ice-rich material can produce CPR values greater than one.

However, rough rocky terrains also generate elevated CPR values. "To reduce ambiguity, another radar diagnostic known as degree of polarisation (DOP) was used. Regions showing CPR values above 1 and DOP values below 0.13 must likely pointed to the presence of subsurface ice," said

Bharti. "The signatures helped identify regions likely to host buried ice deposits."

One crater inside Faustini especially interested researchers as it showed the possibility that the impact may have been on a surface that had subsurface ice in the past, and the impact may have excavated it to this potentially buried ice deposit.

Chandrayaan missions have been instrumental in pointing to evidence of water on the Moon. Anil Bhattacharya, director of PRL and co-author, said that the latest study adds to the knowledge of the potential location where the future missions could look.

The researchers caution that not all craters meet dual radar values but point to more potential sites while searching for scarce water. Both Chandrayaan-2 and -3 targeted the South Pole region because of its scientific importance and potential volatiles deposits.

Expected Outcomes

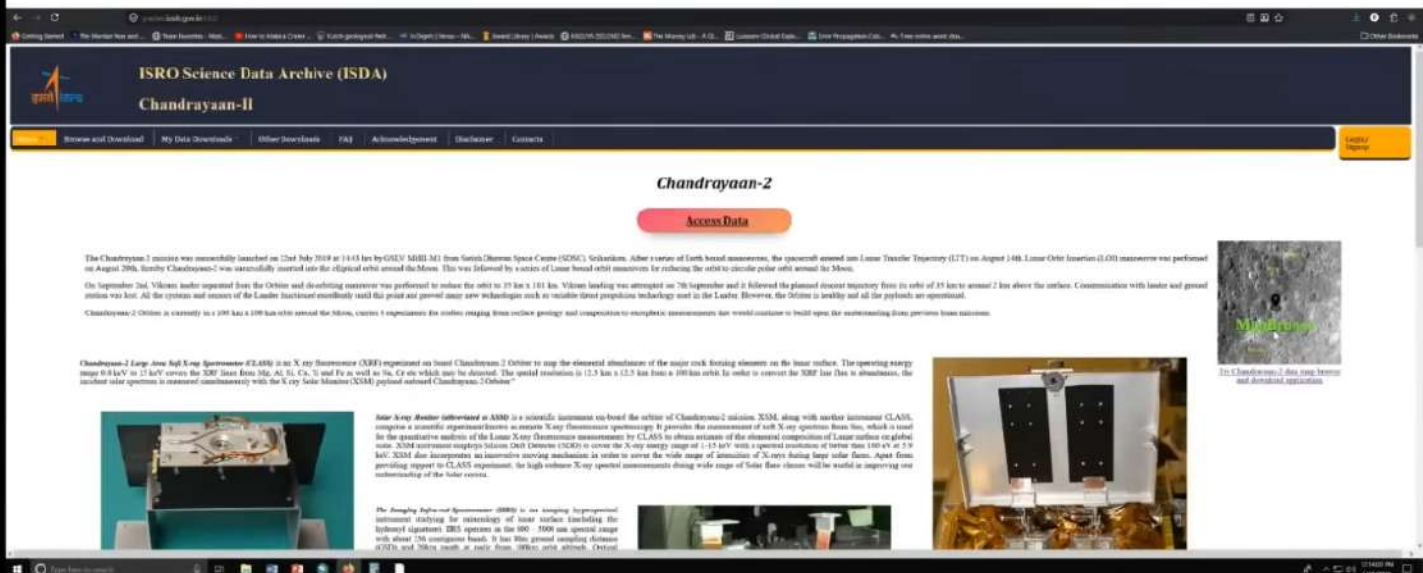
- Identification of *high-probability subsurface ice regions* in doubly shadowed craters
- A *validated radar-based detection framework* for subsurface ice
- A feasible *landing site* near scientifically relevant targets
- An optimized *rover traverse path*
- Quantitative estimates of *subsurface ice volume*



Suggested Tools/Technologies

- GIS platforms: QGIS / ArcGIS
- Programming: Python (NumPy, SciPy, GDAL, rasterio)
- Image processing: ENVI
- **DFSAR Processing: MIDAS**
- Terrain analysis: Digital Elevation Models (DEM tools)
- Path planning: Optimization algorithms / AI-based navigation models
- Visualization: QGIS / ArcGIS / MATLAB / Python-based plotting

Data and Software can be downloaded from:



The screenshot displays the ISRO Science Data Archive (ISDA) website for Chandrayaan-II. The page features a navigation bar with links such as "Browse and Download", "My Data Downloads", "Other Downloads", "FAQ", "Acknowledgement", "Disclaimer", and "Contact". A prominent "Chandrayaan-2" header is followed by an "Access Data" button. The main content area includes a paragraph about the Chandrayaan-2 mission's launch and orbit, a section titled "Chandrayaan-2 Large Area Soft X-ray Spectrometer (LASPEX)" describing its purpose and capabilities, and a section titled "Solar X-ray Monitor (SCM)" detailing its scientific objectives and instruments. The page is visually supported by images of the spacecraft and its instruments.

The Chandrayaan-2 mission was successfully launched on 22nd July 2019 at 14:03 hrs by GSLV MkIII-M1 from Satish Dhawan Space Centre (SDSC), Sriharikota. After a series of Earth based manoeuvres, the spacecraft entered into Lunar Transfer Trajectory (LTT) on August 14th. Lunar Orbit Insertion (LOI) manoeuvre was performed on August 20th, thereby Chandrayaan-2 was successfully inserted into the elliptical orbit around the Moon. This was followed by a series of Lunar based orbit manoeuvres for reducing the orbit to circular polar orbit around the Moon.

On September 2nd, Vikram lander separated from the Orbiter and descending manoeuvre was performed to reduce the orbit to 35 km x 183 km. Vikram landing was attempted on 26 September and it followed the planned descent trajectory from its orbit of 35 km to around 2 km above the surface. Communication with lander and ground station was lost. All the systems and sensors of the Lander functioned nominally until the point just prior to the final engine shutdown to initiate final propulsive technology used in the Lander. However, the Station is healthy and all the payloads are operational.

Chandrayaan-2 Orbiter is currently in a 100 km x 100 km orbit around the Moon, carries 6 experiments for surface imaging, terrain profiling, geology and composition to morphological, geochemical, geophysical measurements that would continue to build upon the understanding from previous lunar missions.

Chandrayaan-2 Large Area Soft X-ray Spectrometer (LASPEX) is an X-ray fluorescence (XRF) experiment on board Chandrayaan-2 Orbiter to map the elemental abundances of the major rock forming elements on the lunar surface. The operating energy range 0.1 keV to 1.5 keV covers the XRF lines from Mg, Al, Si, Ca, Ti and Fe as well as Si, Cr etc which may be detected. The spatial resolution is 12.5 km x 12.5 km from a 100 km orbit. In order to convert the XRF line data to abundance, the incident solar spectrum is measured simultaneously with the X-ray Solar Monitor (XSM) payload onboard Chandrayaan-2 Orbiter.

Solar X-ray Monitor (SCM) is a scientific instrument onboard the orbiter of Chandrayaan-2 mission. SCM, along with another instrument CLASS, comprise a scientific experiment known as solar X-ray fluorescence spectroscopy. It provides the measurement of soft X-ray spectrum from Sun, which is used for the quantitative analysis of the Lunar X-ray fluorescence measurements by CLASS to obtain estimates of the elemental composition of Lunar surface or global scale. SCM instrument employs Silicon Carbide Diodes (SCDs) to cover the X-ray energy range of 1-15 keV with a spectral resolution of better than 100 eV at 5.9 keV. SCM also incorporates an innovative shielding mechanism in order to cover the wide range of intensities of X-rays during large solar flares. Apart from providing support to CLASS experiment, the high volume X-ray spectral measurements during wide range of Solar flare events will be useful in understanding our understanding of the solar events.

The **Highly Sensitive Spectrometer (HSS)** is an imaging spectrograph instrument studying the intensity of solar surface (including the hydrogen spectrum) RES operates in the 100 - 5000 Å spectral range with about 100 channels bands. It has fine grained sampling domain (0.75 Å) and 3000 pixels in x , y plane from -1000 to 1000 pixels.

ISDA Chandrayaan-2 data, map, images and derived operations

<https://pradan.issdc.gov.in/ch2/>

Expected Solution / Steps to be followed to achieve the objectives

- Map permanently shadowed regions and identify *doubly shadowed craters* using illumination models and OHRC imagery.
- Analyze *DFSAR data* to compute radar parameters such as Circular Polarization Ratio (CPR) and Degree of Polarization (DOP). Apply refined criteria (e.g., $CPR > 1$ and $DOP < 0.13$) to identify potential subsurface ice signatures.
- Study crater morphology, slopes, boulder distribution, and surface roughness using *OHRC data*.
- Evaluate terrain safety and proximity to *ice-bearing regions*.
- Design an *optimal and safe path* considering terrain hazards and solar power constraints.
- Use radar backscatter models and dielectric assumptions to estimate ice concentration and volume within the top 5 meters.

meters.

In this Challenge #8, we are asking you to think like mission planners. Your task is to move from orbital observations to an actionable exploration strategy: where to land, where to drive, where to excavate, and how much ice may be available.

Your challenge is to discover hidden lunar ice and design a realistic pathway for future ISRO missions to reach and utilize it.